# M7 — What this block gets you

Lake, warehouse, lakehouse: the vocabulary of your own architecture doc.

- The industry words for what you are already building— lake, warehouse, lakehouse
- Why warehouses alone and lakes alone both fail— and why the lakehouse won
- Every term mapped to your target design— SQL Server DW → BigQuery, Bronze/Silver/Gold
- Why 'open' formats (Parquet, Iceberg) are your insurance against lock-in
- The deck's built-in quiz, run as a group— if you can answer it, you can defend the design doc

# Day 2, first block — 09:00 to 10:30

| | | |
|---|---|---|
| **09:00–09:15** | Recap quiz — which GCP tool replaces which piece of your current stack? | QUIZ |
| **09:15–09:45** | M7 — Intro to modern DE: lake vs warehouse vs lakehouse (deck quiz pages at the end run as a group) | |
| **09:45–10:30** | M8 — Building a data lakehouse: GCS, open formats, Iceberg, federated queries | |
| **10:30–10:45** | Break | |

QUIZ

# Which GCP tool replaces which piece of your current stack?

15 minutes · Whole room — confer at your table

1. Instructor reads each question aloud — 10 questions, no trick ones
2. Confer with your table for about 30 seconds per question
3. One person answers for the group — nothing is scored
4. Name, out loud, what each piece of your SQL Server stack becomes on Google Cloud
5. Jot your answers down — the key is on the next slide

Why it matters — retrieval practice before the lakehouse modules start — proving Day 1 stuck, in your own vocabulary, not Google's.

# Quiz key — your stack, translated

The short version of yesterday; the deck quiz at the end of this module checks the new vocabulary.

- Nightly extract jobs— Datastream change data capture, replicating continuously
- SQL Server Agent— Cloud Scheduler; cheapest version is a BigQuery scheduled query
- SSIS packages— Dataform for SQL-first ELT; Dataflow only when SQL is not enough
- Staging file drops— land in Cloud Storage (Bronze), load via bq load, LOAD DATA, or DTS
- Linked servers— federated queries and external tables, no copy, always current
- SSAS cubes— materialized views and Gold marts, built and scheduled by Dataform

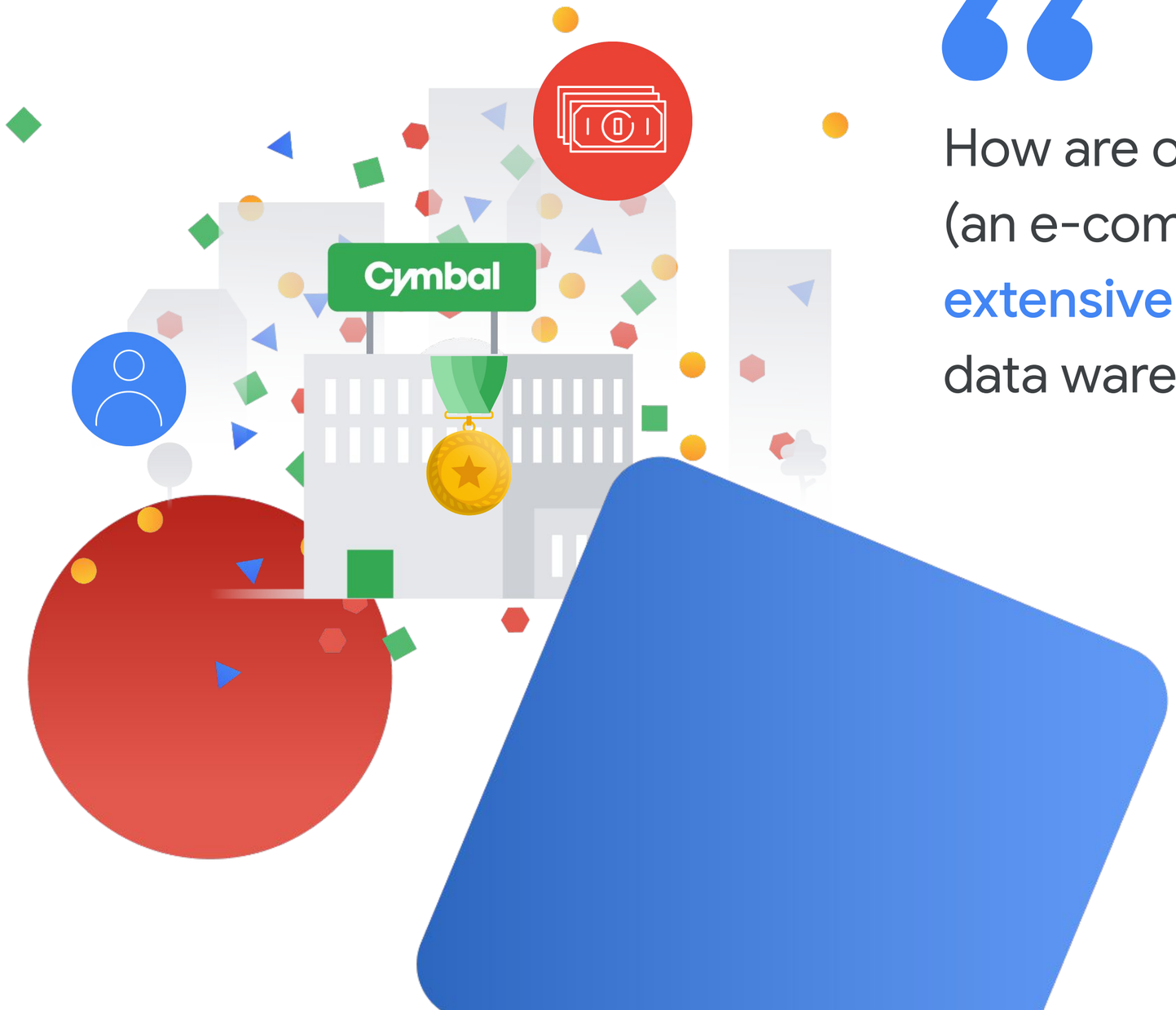
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For years, organizations have relied on two primary methods for large-scale data storage: data lakes and data warehouses.

# This is the vocabulary of your own architecture doc.

Frame for the next 30 minutes — lake, warehouse and lakehouse are the words in the DOM DoIT target design, not textbook terms.

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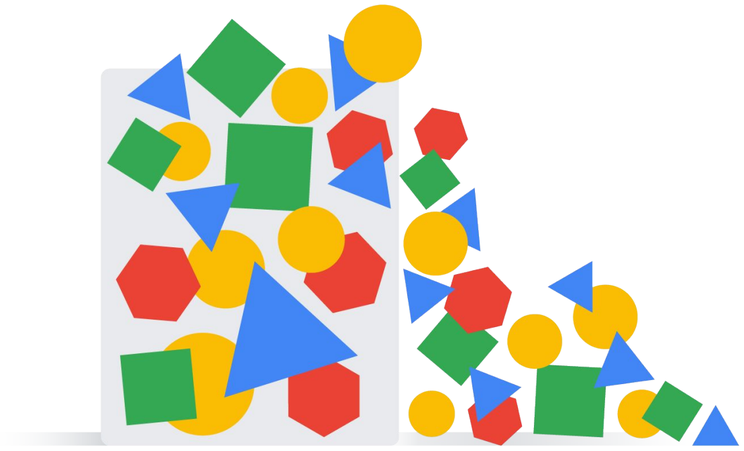



How are organizations, such as Cymbal (an e-commerce company), managing extensive datasets with data lakes and data warehouses?

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# Data lakes

Think of a data lake as a vast reservoir of data. It stores enormous amounts of raw data in its native format.



01 Structured data

02 Semistructured data

03 Unstructured data

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# Data lakes

01

Data lakes operate on the principle of "store everything now, figure out how to use it later."

02

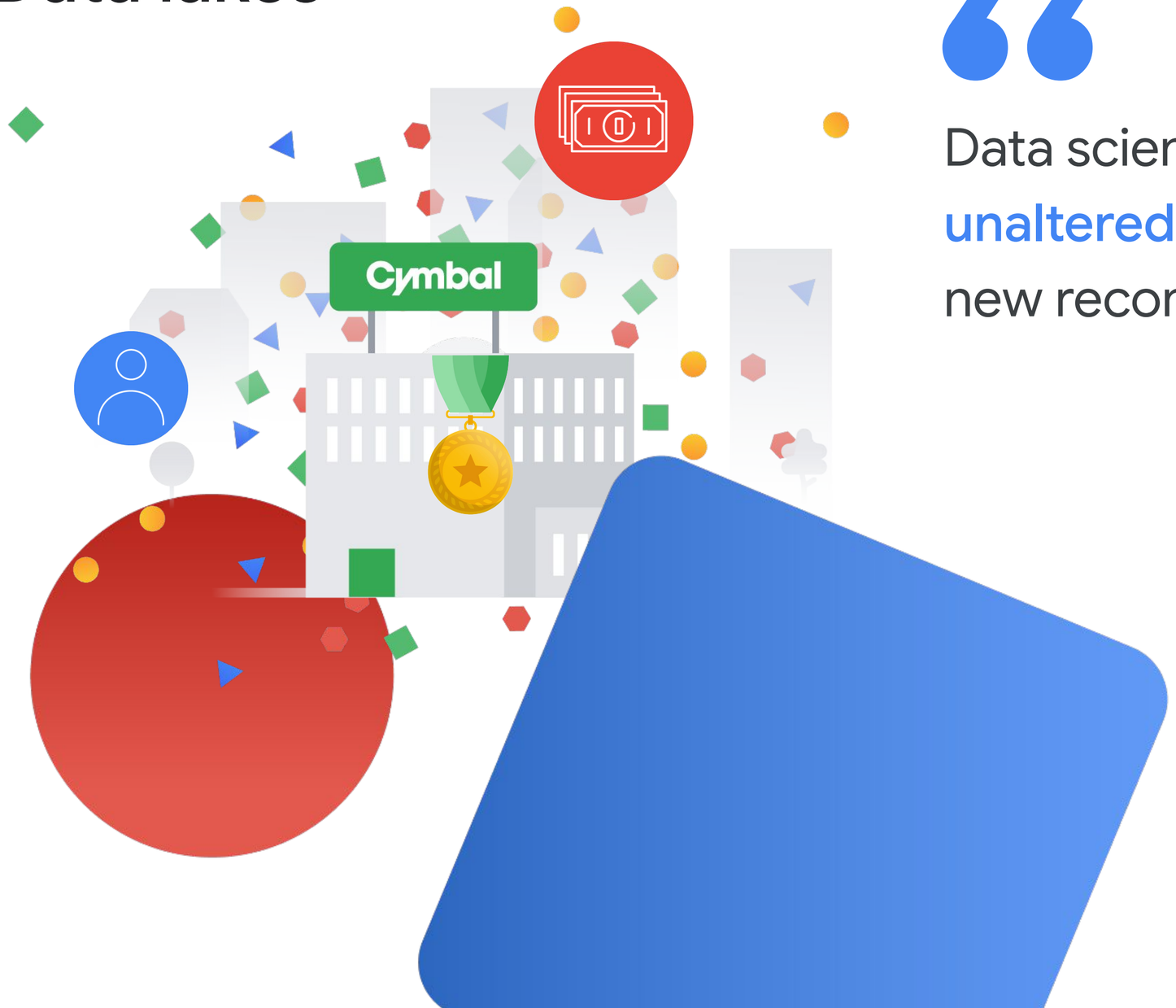
This approach applies a structure to data upon reading (schema-on-read), not during storage.

03

Data lakes are well suited for data exploration, machine learning (ML), and big data processing.

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# Data lakes




Data scientists need access to original, **unaltered clickstream data** to build a new recommendation engine.

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# Data lakes: Advantages

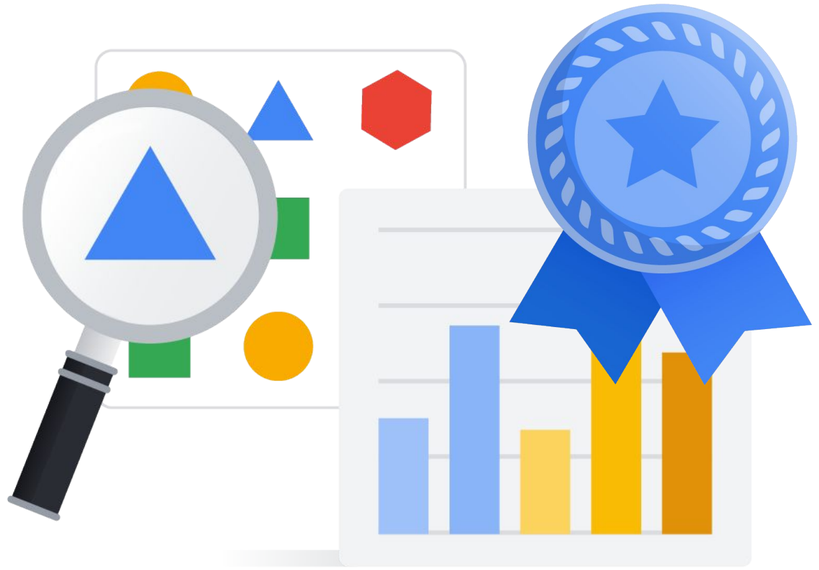
01 **Flexibility:** Stores all data types.

02 **Agility:** Fast to ingest data.

03 **Scalability:** Can grow to exabyte scale.

04 **Cost-effectiveness:** Uses low-cost object storage.

05 **Support for advanced analytics:** Ideal for AI/ML model training.



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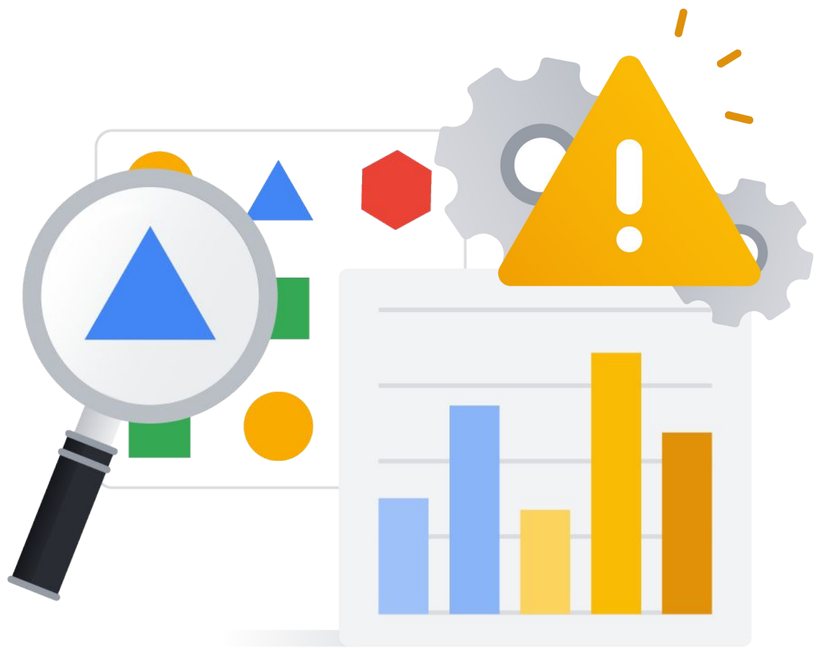
# Data lakes: Disadvantages

01 **Risk of becoming a "data swamp":** Can become a disorganized collection of unusable data.

02 **Management complexity:** Requires significant overhead to maintain.

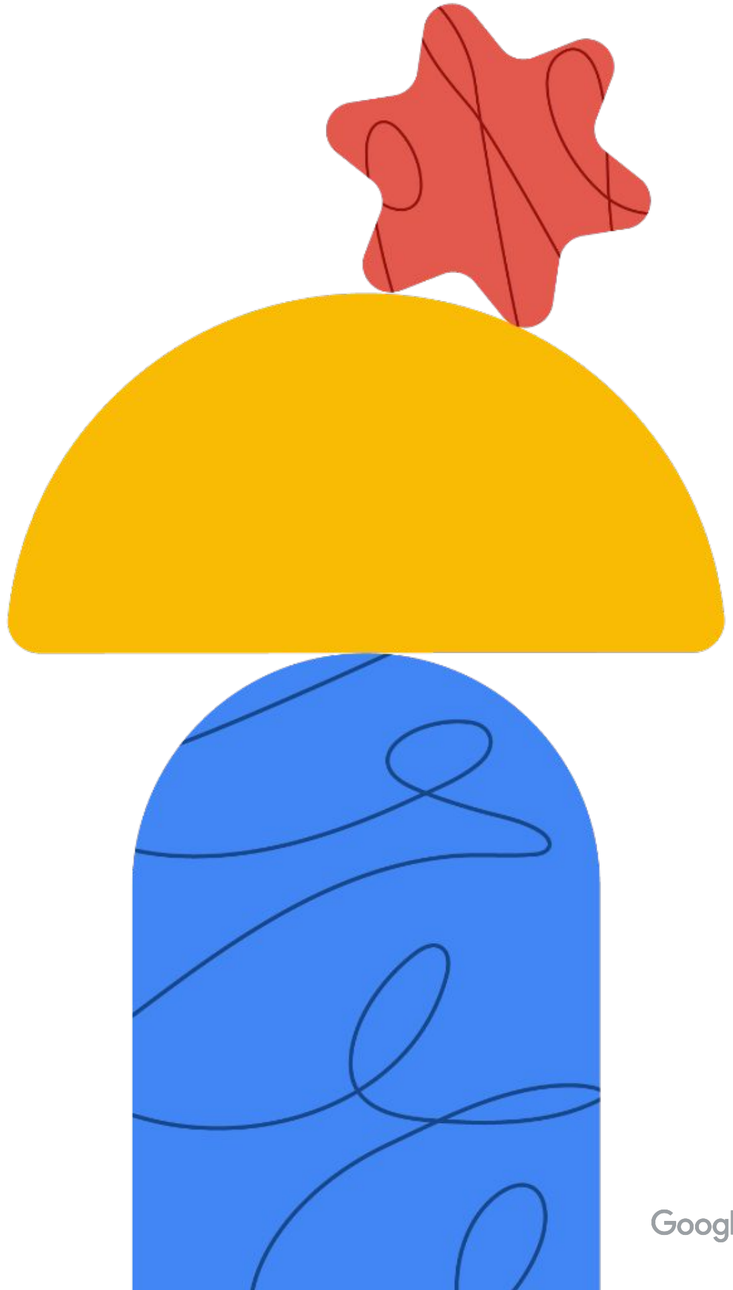
03 **Time-consuming analysis:** Data needs to be cleaned and wrangled before it is usable.

04 **Security risks:** Raw data formats can increase security and compliance risks.



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A data warehouse is like a highly organized library.

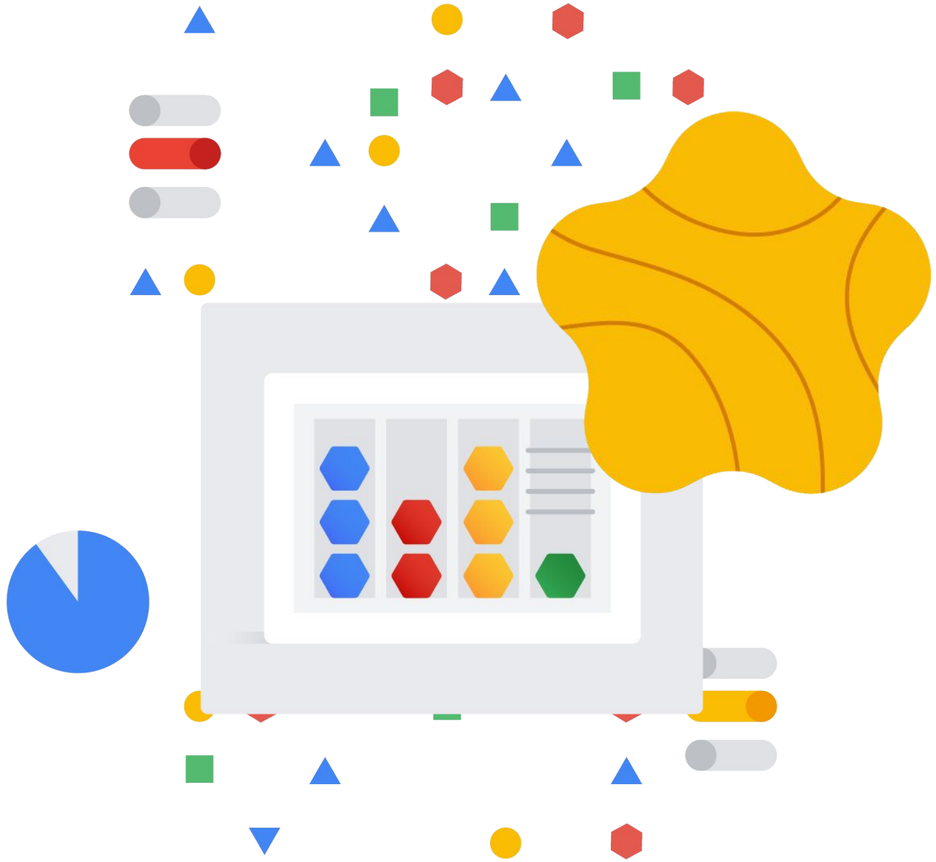


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# Data warehouses

01 Data is cleaned, transformed, and structured before storage.

02 This process, schema-on-write, optimizes data for analysis and business intelligence (BI).



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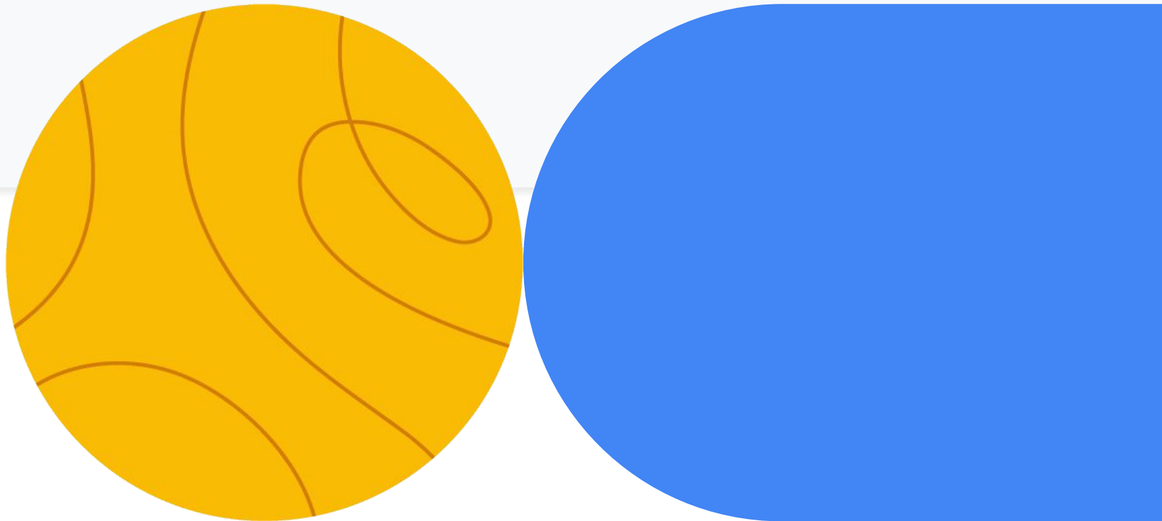
# Data warehouses

## Structure and quality first

The intended use of the data must be determined before the data warehouse is designed.

## Cymbal example

- Key metrics, such as customer lifetime value and daily sales figures, are already defined.
- The data warehouse is then built to provide fast and accurate reports on these specific metrics.



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Data warehouses are perfect for generating business reports and dashboards where speed and accuracy are critical.

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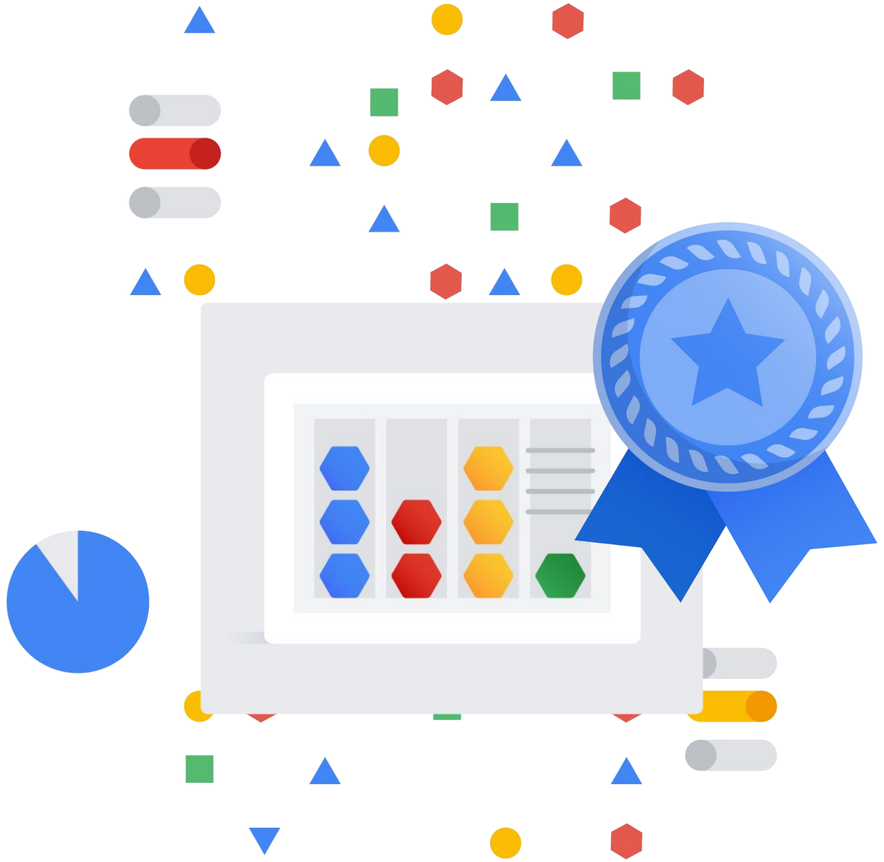
# Data warehouses: Advantages

01 **Speed:** Optimized for high-performance queries.

02 **High-quality data:** Provides consistent and reliable information.

03 **Business focused:** Directly answers common business questions.

04 **Historical intelligence:** Enables deep analysis of trends over time.



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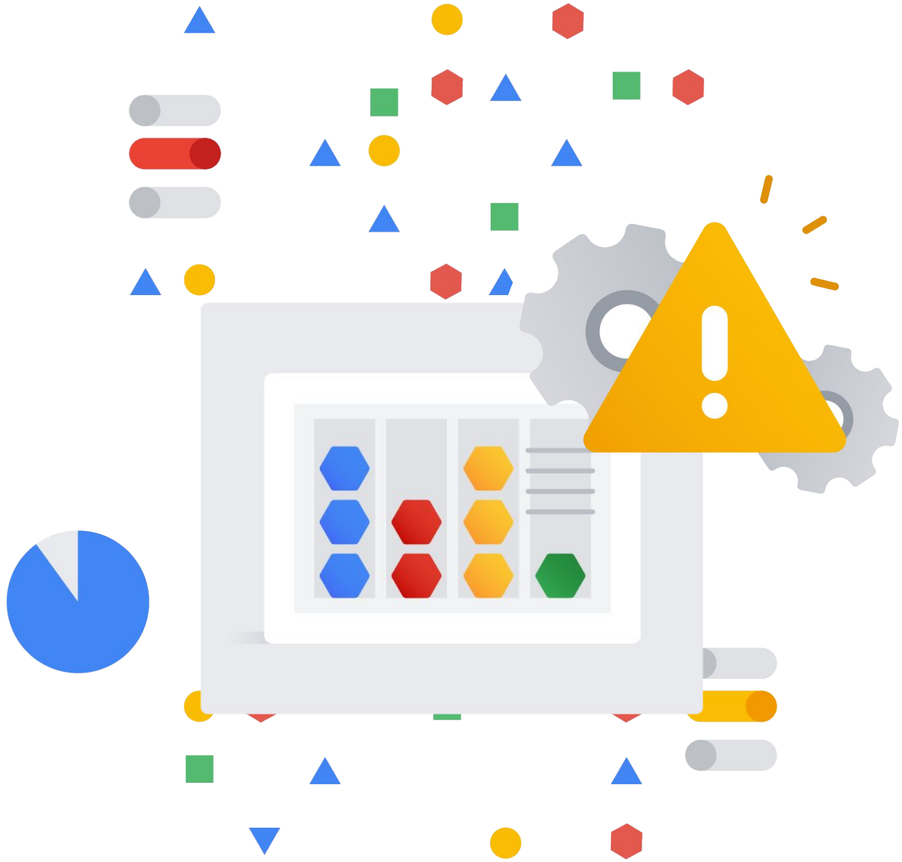
# Data warehouses: Disadvantages

01 **Inflexibility:** Cannot easily accommodate new data types or unstructured data.

02 **Cost:** Often expensive to build and maintain.

03 **Limited data types:** Primarily designed for structured data only.

04 **Long development time:** Can take a long time to design and implement.



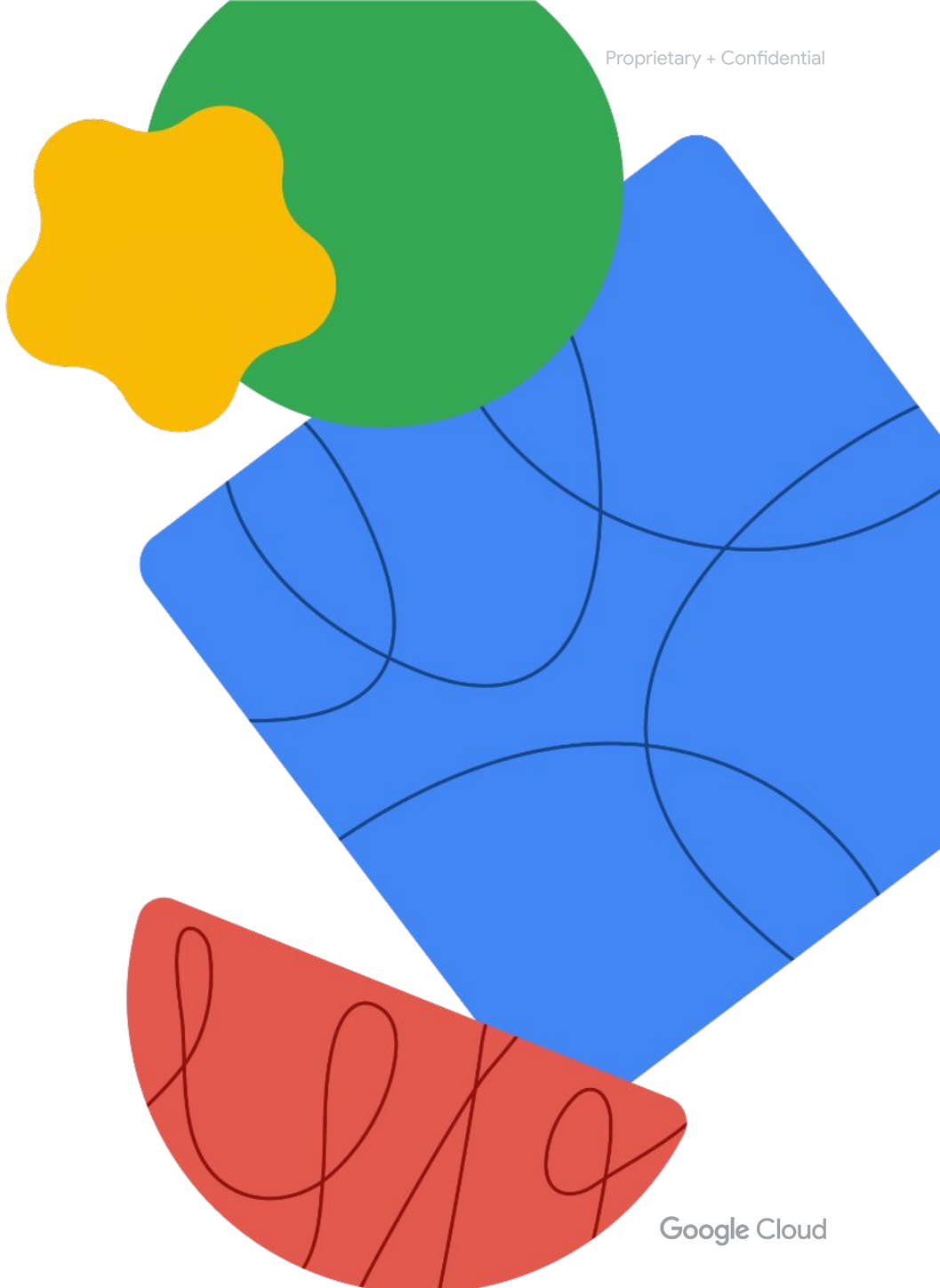
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Neither data lakes or data warehouses excel at supporting the full range of an organization's data needs.

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# The modern approach: Data lakehouse

Combines the flexibility and low cost of a data lake with the speed and accuracy of a data warehouse.



# Lake, warehouse, lakehouse — in your project's terms

Your target design IS a lakehouse design; this module gives you the words to defend it.

- The lake— Cloud Storage buckets; your Bronze layer, raw data landed as-is
- The warehouse— BigQuery; your Silver and Gold, cleaned and business-ready
- The lakehouse— the two acting as one system, BigQuery querying both
- Schema-on-write— your world today: everything modeled before it lands
- Schema-on-read— land raw in Bronze first, model into Silver/Gold when ready
- One engine owning the data— replaced by storage and compute scaled independently

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# Data lakehouse

01

## Unified platform

Combines low-cost storage of a data lake with data warehouse management and query performance.

02

## Versatile workloads

Supports traditional BI, modern data science, and AI without data movement or duplication.

03

## Optimized architecture

Implements metadata and governance on open-format files in low-cost object storage (such as Google Cloud Storage) for a hybrid approach.

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# Data lakehouse

## Data silo challenges

Cymbal’s sales data was in a data warehouse and its customer review data was in a data lake. The two could not be analyzed together easily.

## Integrated data analysis

With a data lakehouse, Cymbal can now run a single query to correlate customer sentiment from text reviews with sales trends from its transaction database.

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# Data lakehouse: Key features

01 Support for most data formats

02 Flexible schema-on-read or schema-on-write

03 Access for all types of data users

04 Cost flexibility based on needs

05 Unified data governance

06 ACID transaction support

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# Lakehouse technology

- Google Cloud uses Lakehouse to implement data lakehouses.
- Lakehouse extends Google Cloud's governance and query features to data in Google Cloud or other major clouds.
- Manage and query data in its original location using open formats, such as Parquet, ORC, and Avro.
- Utilize BigQuery's powerful query engine.

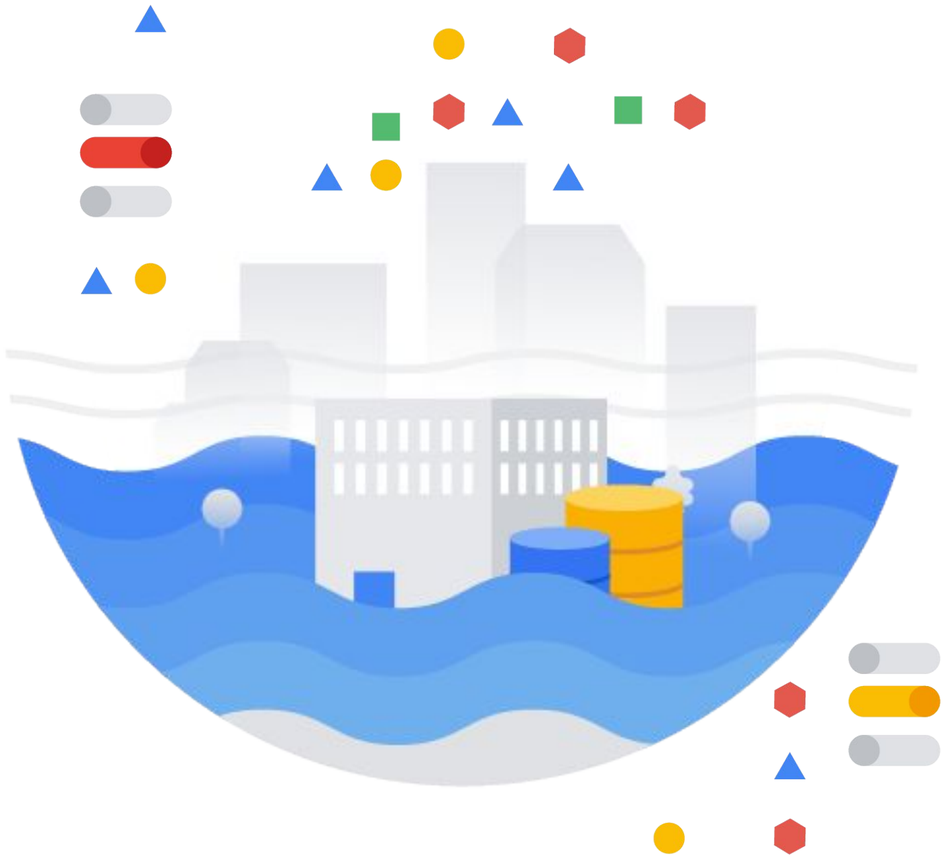
# 'Open' is the load-bearing word

- Open file formats— Parquet, ORC, Avro: no single engine owns your bytes
- Open table formats— Apache Iceberg adds ACID, schema evolution, time travel to files
- Your Bronze files stay readable by non-Google engines— insurance against lock-in
- Multiple engines— BigQuery, Spark, others — work safely on the same tables
- This is why the Bronze decision (Parquet on GCS, not CSV) was yesterday's real lesson

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# Data lakehouse benefits

- Unified data storage and access
- Reduced data redundancy
- Unified governance
- Broken-down data silos
- Improved flexibility and scalability



QUIZ

# Built-in quiz — run it as a group

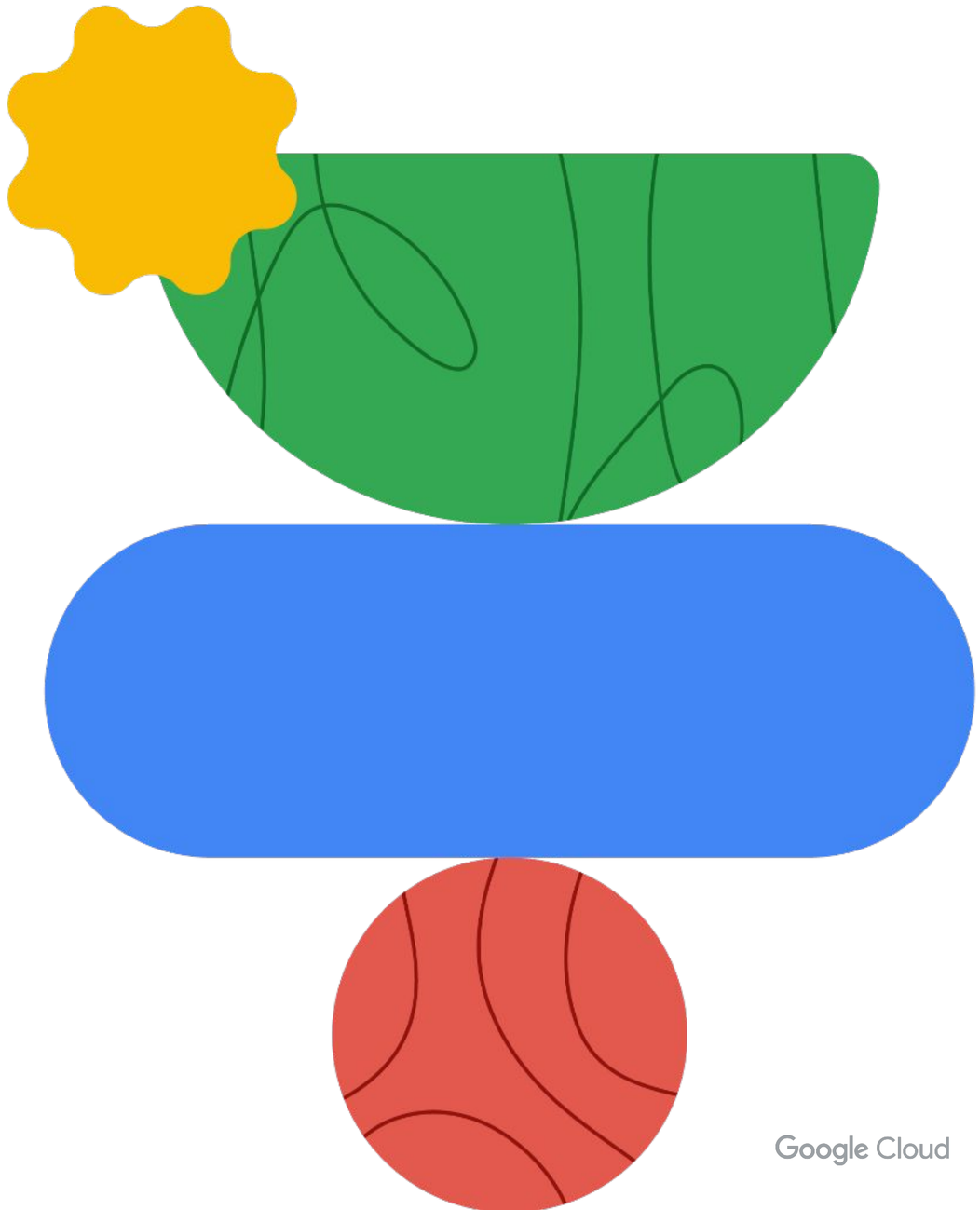
Last 5–8 minutes of the block · Whole room — the quiz pages that follow in this deck

1. Instructor shows each question page and reads it aloud
2. Confer at your table for about 30 seconds, then answer for the group
3. Treat wrong answers as vocabulary gaps to fix now, not as grades
4. Four questions with answer pages — lake, warehouse, lakehouse, and the Cymbal scenario

Why it matters — if you can answer this quiz out loud, you can narrate your own architecture doc to anyone who asks 'why not just a warehouse?'

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# Quiz



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# Quiz | Question 01

## Question

Which of the following best describes the primary characteristic of a data lake?

A. Data is cleaned, transformed, and structured before storage, optimized for BI.

B. It is a highly organized library designed for fast and accurate reports based on predefined metrics.

C. It stores enormous amounts of raw data in its native format with schema-on-read, ideal for exploration and ML.

D. It primarily handles structured data and is optimized for high-performance queries on consistent information.

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# Quiz | Question 01

## Answer

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# Quiz | Question 02

## Question

What is a key disadvantage of a traditional data warehouse?

A. It risks becoming a "data swamp" without proper governance.

B. It is inflexible and cannot easily accommodate new data types or unstructured data.

C. It primarily uses low-cost object storage, leading to security risks with raw data.

D. It is ideal for AI/ML model training due to its support for advanced analytics.

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# Quiz | Question 02

## Answer

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# Quiz | Question 03

## Question

How does a data lakehouse architecture combine the best features of data lakes and data warehouses?

A. By enforcing a strict schema-on-write approach for all data types to ensure high data quality.

B. By creating separate, specialized platforms for BI and AI workloads to maximize performance.

C. By implementing a metadata and governance layer on top of open-format files in low-cost object storage.

D. By always requiring data to be moved and duplicated between a data lake and a data warehouse.

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# Quiz | Question 03

## Answer

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# Quiz | Question 04

## Question

Cymbal, an e-commerce company, needs a solution to support traditional BI, modern data science, and AI workloads on a single, governed copy of its data while also breaking down data silos between sales and customer review data. Which architecture is best suited for this need?

A. A data warehouse, such as BigQuery, for high-speed, interactive BI on structured data.

B. A data lake, using Cloud Storage, for initial, low-cost storage of massive volumes of raw data for AI/ML research.

C. A data lakehouse architecture with BigQuery and Lakehouse.

D. Separate data lakes and data warehouses, with manual data integration as needed.

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# M7 — What you now have

- The lake/warehouse/lakehouse vocabulary— and why the lakehouse collapses both silos into one platform
- Your design in those terms— GCS is the lake (Bronze), BigQuery the warehouse (Silver/Gold), one system
- The case for open formats— Parquet and Iceberg keep your Bronze free of lock-in
- The group quiz answered— vocabulary you can reuse when you defend the architecture doc
- Next: M8 builds the lakehouse itself— storage options, Iceberg, federated queries (09:45–10:30)